

Computer Networks – Presentation Speaking Script

Slide 1 – Title: Network Foundations

Hello everyone. My name is Arya Desai, and today I will be presenting on computer network foundations.

In this presentation, I will explain what computer networks are, how the network edge and network core work, how protocols help devices communicate, how network performance is measured, and why network security is important.

I will also show a short practical demonstration using the ping command.

Slide 2 – Network Foundations

A computer network is a group of connected devices that exchange data and share resources.

These devices can include laptops, smartphones, servers, Internet of Things devices, and cloud systems.

The Internet is often described as a network of networks because many smaller networks are connected together.

For communication to work, networks need devices, links, addressing, and common protocols.

In everyday life, I use networks when I connect my Mac to my iPhone hotspot or when I access Blackboard and other websites for coursework.

Slide 3 – Network Edge and Network Core

A network can be understood in two main parts: the network edge and the network core.

The network edge includes the devices that send and receive information. These devices are called hosts.

A client usually requests a service, while a server provides that service.

Access networks connect users and their devices to the larger network. Examples include home Wi-Fi, business networks, and mobile networks.

The network core is responsible for moving data between different networks. Routers forward packets, switches connect devices inside local networks, and links carry the data.

The main idea is that the edge creates or receives the data, while the core helps deliver it.

Slide 4 – Network Diagram

This diagram shows a simple example of how a student's device can communicate with a web server.

The process begins with a student using a laptop connected to a home Wi-Fi network.

The data then reaches a router and enters the Internet, which is the network core.

From there, the packet may pass through several routers and links before reaching the destination web server.

The server sends a response back through the network toward the student's laptop.

This shows that communication usually involves several devices and network links rather than one direct connection.

Slide 5 – How Data Travels Through the Core

When a message is large, it is divided into smaller units called packets.

Routers examine information in each packet and forward it toward the destination.

Packet switching allows many users to share the same network links. This makes better use of the available network capacity.

Packets belonging to the same message may experience different delays or even different paths.

This is different from circuit switching. In circuit switching, a dedicated path or capacity is reserved for a communication session.

The Internet mainly uses packet switching because it is flexible and allows network resources to be shared.

Slide 6 – Communication and Network Protocols

Protocols are rules that allow different devices and systems to communicate.

DNS translates a website name, such as google.com, into an IP address.

IP provides addressing and helps packets move between networks.

TCP supports reliable and ordered delivery between applications.

HTTP and HTTPS are used for communication between a web browser and a web server. HTTPS also provides protection through encryption.

Protocols are important because devices need shared rules for formatting data, addressing, timing, and handling information.

Slide 7 – Browser Scenario: Entering a Website Address

This slide shows how several protocols work together when someone enters a website address into a browser.

First, DNS finds the IP address connected to the website name.

Next, IP helps address the packets and move them toward the correct server.

TCP provides reliable transport between the communicating endpoints.

Finally, HTTPS is used to request and receive web content securely.

The important point is that these protocols do not work separately. They cooperate to complete one simple action, such as opening a website.

Slide 8 – Understanding Network Performance

Network performance describes how quickly and reliably data moves through a network.

Bandwidth is the capacity of a link. It describes how much data the link can carry.

Throughput is the actual rate of successful data transfer.

Delay is the time needed for data to travel. It can include transmission delay, which is the time required to place bits onto a link, and queuing delay, which is the time data waits in a router or switch.

Packet loss happens when packets do not reach their destination.

Congestion can increase delay and packet loss. Distance, traffic, device performance, and link capacity can also affect network performance.

That is why actual measurements are useful when evaluating a network.

Slide 9 – Network Demonstration: Ping

For the practical demonstration, I used the macOS Terminal and ran this command:

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ping -c 4 google.com
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The command sends four test packets to google.com and waits for replies.

The destination resolved to the IP address 142.251.41.142.

Four packets were sent, and all four were received. This resulted in zero percent packet loss.

The minimum round-trip time was 7.630 milliseconds, the average was 32.840 milliseconds, and the maximum was 74.686 milliseconds.

The response times were different, which means that latency changed during the test.

Overall, the test was successful because the destination responded to every packet.

Slide 10 – Terminal Evidence: Ping Results

This slide shows the actual terminal screenshot from the ping test.

The screenshot confirms that four packets were transmitted and four packets were received.

It also shows zero percent packet loss and the minimum, average, maximum, and standard deviation values.

The third reply had the highest latency in the test. This is a simple example of how network response time can vary from one packet to another.

The ping command is useful for basic connectivity testing, although it does not measure every aspect of network performance.

Slide 11 – Network Security

Network security focuses on protecting communication, devices, and information.

Authentication verifies the identity of a user or device.

Encryption changes readable information into protected data so that unauthorized people cannot easily understand it.

Confidentiality means that information should only be available to authorized people.

Some common threats include unauthorized access, malware, IP spoofing, and packet sniffing.

IP spoofing happens when an attacker falsifies the source IP address in a packet.

Packet sniffing involves capturing and examining network traffic.

These threats show why security must be considered whenever devices communicate over a network.

Slide 12 – Security Strategies and Real-World Scenario

A common real-world example is a student using public Wi-Fi in a café or airport.

An attacker may try to observe network traffic or pretend to be another device.

There are several ways to reduce these risks.

Users should prefer HTTPS and encrypted connections, use strong and unique passwords, and enable multi-factor authentication.

A trusted VPN may be useful in some situations. Wi-Fi networks should use strong encryption, and operating systems and applications should be kept updated.

Firewalls and network monitoring can also help.

Users should verify the correct Wi-Fi network name and avoid entering sensitive information on unknown or suspicious networks.

Slide 13 – Conclusion and References

To conclude, computer networks are more than just cables and devices.

They are coordinated systems that move information through end devices, routers, links, protocols, and security controls.

In this presentation, I explained network foundations, the network edge and core, packet switching, protocols, network performance, the ping demonstration, and basic security practices.

The practical test helped connect the theory to a real network measurement.

Thank you for listening. I am happy to answer any questions.